



Introduction to Photogrammetry

Lecture 17

Orientation of a Stereopair

Usman Mazhar, Ph.D.

usmanmazhargeo@outlook.com

In Today's Class

1. Relative Orientation
2. Absolute Orientation

1. Relative Orientation

- Relative orientation determines the orientation of one photograph with respect to another photograph.
- The process reconstructs the geometric relationship between overlapping images.
- Relative orientation allows the formation of a three-dimensional stereo model.
- It is an important stage before absolute orientation and mapping.

- Relative orientation removes y-parallax between overlapping photographs.
- The process reconstructs the original imaging geometry of stereo pairs.
- It enables accurate stereoscopic viewing and terrain interpretation.
- Relative orientation forms the basis of three-dimensional photogrammetric measurement.

Geometry of Relative Orientation

- Relative orientation reconstructs the geometric relationship between two perspective centers.
- The orientation process uses conjugate image points visible in both photographs.
- Geometric conditions are mathematically modeled using orientation equations.
- Stereo geometry supports terrain reconstruction and mapping.

Coplanarity Condition

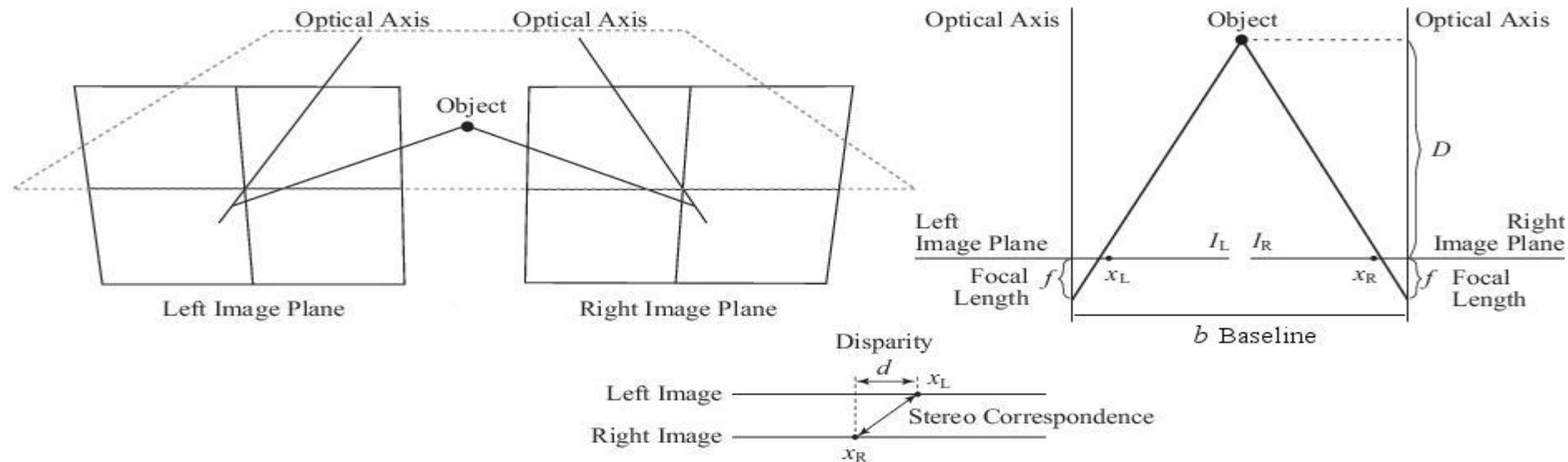
- The perspective centers and image rays of corresponding points must lie in the same plane.
- This requirement is known as the coplanarity condition.
- Coplanarity forms the mathematical basis of relative orientation.
- Violations of coplanarity indicate orientation inconsistencies.

Relative Orientation Parameters

- Relative orientation requires the determination of angular orientation parameters.
- These parameters describe the rotation between overlapping photographs.
- Relative orientation also establishes the baseline relationship between images.
- The computed parameters support stereo model formation.

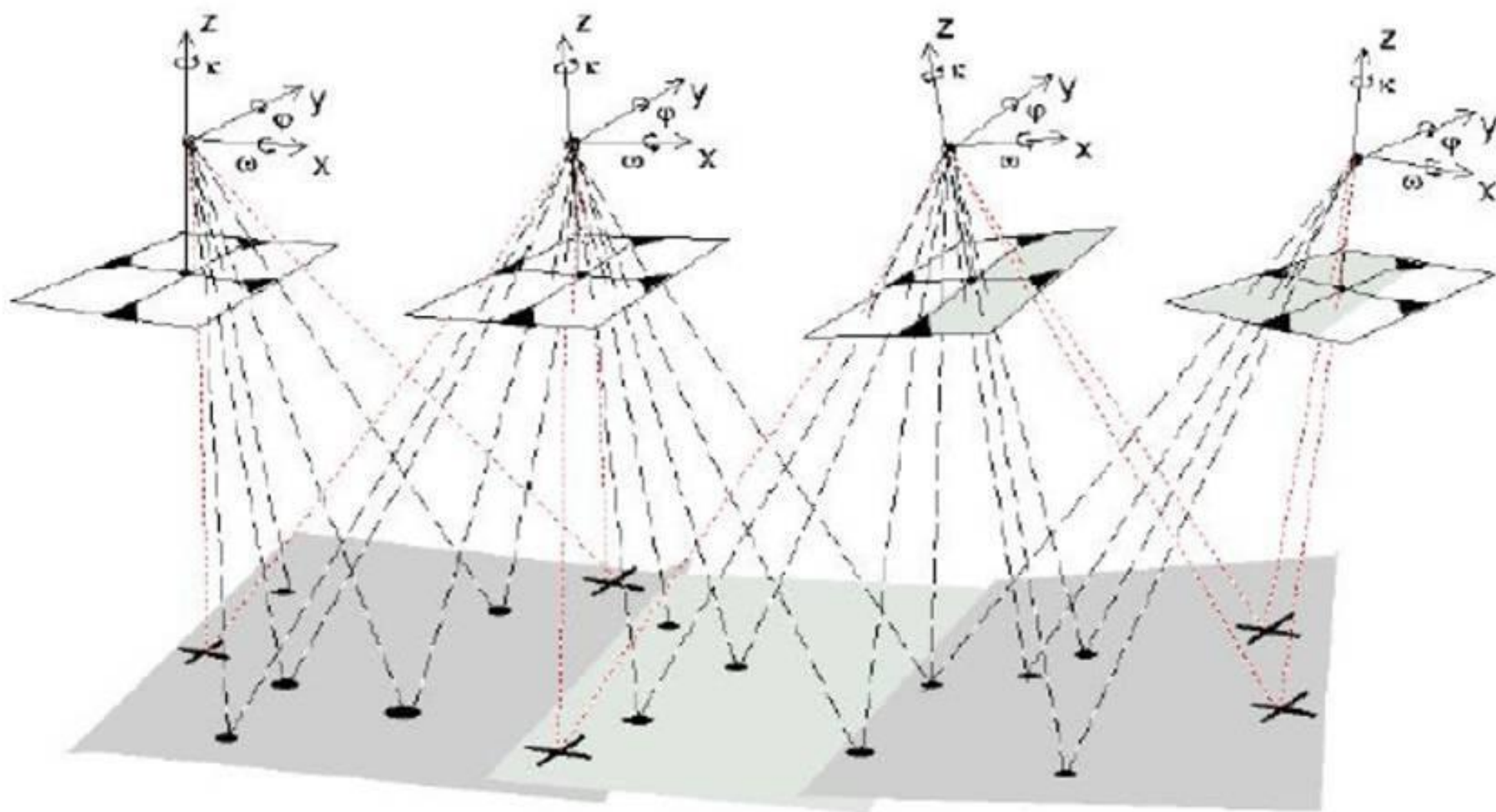
Baseline Concept

- The baseline is the distance between the two camera exposure stations.
- Baseline length influences stereo coverage and elevation accuracy.
- Longer baselines generally improve height measurement sensitivity.
- Baseline geometry is important in stereo model reconstruction



Tie Points in Relative Orientation

- Tie points are common points identified in overlapping photographs.
- These points connect the geometry of stereo image pairs.
- Tie points are used to compute orientation parameters mathematically.
- Accurate tie point selection improves orientation reliability.



Manual Relative Orientation

- Traditional photogrammetry used manual stereoscopic instruments for orientation.
- Operators adjusted image positions until y-parallax disappeared visually.
- Manual methods required highly skilled photogrammetrists.
- Analytical methods later improved precision and automation.

Analytical Relative Orientation

- Analytical photogrammetry uses mathematical equations for orientation computation.
- Computer algorithms determine orientation parameters automatically.
- Analytical methods provide higher accuracy than manual procedures.
- Modern software performs relative orientation efficiently.

Digital Relative Orientation

- Digital photogrammetry uses automated image matching techniques.
- Computer vision algorithms identify tie points automatically.
- Digital processing reduces manual workload significantly.
- Automated orientation supports large-scale mapping projects

Errors in Relative Orientation

- Inaccurate tie points may distort stereo geometry.
- Poor image quality can affect orientation precision.
- Lens distortions and measurement errors introduce uncertainties.
- Proper adjustment procedures reduce orientation errors.

2. Absolute Orientation

- Absolute orientation is the final step in stereo photogrammetric processing that places a 3D model into a real-world coordinate system.
- It transforms a relatively oriented stereo model into a geographically correct and scaled representation.
- The process ensures that the model matches the ground coordinate system such as UTM or WGS84.
- It uses known Ground Control Points (GCPs) that are identifiable in both image space and ground space.
- Without absolute orientation, the stereo model remains only a scaled but non-georeferenced representation.

- The main purpose of absolute orientation is to correctly position the stereo model in the Earth's coordinate system.
- It ensures that the model has the correct scale corresponding to real-world distances.
- It aligns the model's axes with the geographic coordinate system used in mapping.
- It allows integration of photogrammetric output with GIS and remote sensing datasets.
- It converts a locally defined model into a usable spatial dataset for mapping and analysis

Input Data for Absolute Orientation

- Absolute orientation requires a relatively oriented stereo model obtained from the previous processing stage.
- It requires Ground Control Points (GCPs) with known real-world coordinates (X, Y, Z).
- These GCPs must be clearly identifiable in the stereo images for accurate matching.
- A minimum of three non-collinear GCPs is required, although more points improve accuracy.
- Both image/model coordinates and ground coordinates are used simultaneously in the transformation.

Ground Control Points (GCPs)

- Ground Control Points are fixed reference points on the Earth with known coordinates.
- They are selected so that they are clearly visible and accurately measurable in the images.
- GCPs act as a bridge between the image-based model and the real-world coordinate system.
- The accuracy of absolute orientation heavily depends on the quality and distribution of GCPs.
- Poorly distributed or inaccurate GCPs can introduce geometric distortions in the final model.

Mathematical Concept

- Absolute orientation is based on a 3D similarity transformation between model and ground coordinates.
- The transformation includes three translations along X, Y, and Z axes.
- It also includes three rotations around the axes, commonly denoted as ω (omega), φ (phi), and κ (kappa).
- A uniform scale factor is applied to adjust the size of the stereo model to real-world scale.
- Together, these parameters form a seven-parameter transformation model

Transformation Parameters

- The translation parameters shift the model from its local origin to the ground coordinate system origin.
- The rotation parameters align the orientation of the stereo model with the Earth-based coordinate axes.
- The scale factor adjusts differences between model units and real-world measurement units.
- These parameters are computed using mathematical adjustment techniques such as least squares.
- The goal is to minimize the difference between transformed model points and actual ground points.

Computational Procedure

- The coordinates of GCPs in the model are matched with their corresponding ground coordinates.
- A system of equations is formed based on the 3D similarity transformation model.
- Least squares adjustment is used to estimate the best-fit transformation parameters.
- Residual errors are analyzed to ensure that the transformation is geometrically consistent.
- Once computed, the parameters are applied to transform all model points into ground space.

Output of Absolute Orientation

- The output is a fully georeferenced stereo model aligned with real-world coordinates.
- All model points are converted into a consistent coordinate system such as UTM or local grid systems.
- The resulting model has corrected scale, position, and orientation relative to the Earth.
- The output can be directly used for mapping, GIS integration, and spatial analysis.
- This stage completes the photogrammetric workflow from images to usable spatial data.

Importance and Applications

- Absolute orientation is essential for converting photogrammetric models into usable map products.
- It enables integration with GIS layers such as land use, elevation, and infrastructure data.
- It is widely used in topographic mapping, urban planning, and environmental studies.
- It ensures that measurements taken from the model correspond to real-world distances and heights.
- It is a critical step in producing accurate Digital Elevation Models (DEMs) and orthophotos.



Thank you!

Any Questions
